Largen Animal Surgery and Practice

Presurgical consideration & anesthesia



History of Animal Surgery

Ancient Egypt and Ancient India

Military horses and productive animals

- practices were targeted at horses and cattle, which held immense military and agricultural value.

18th Century

Systematization and Progressing alongside Human Medicine

- Veterinary medicine became a modern science.
- Anesthesia and asepsis
- Focused on large animals (cattle, pigs, horses)

Modern Veterinary Surgery

Companion animals

- Following the Industrial Revolution, the agriculture was profoundly impacted, driving the focus of veterinary medicine to shift from farm animals to companion animals.

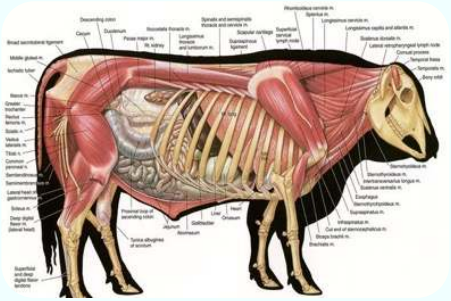
Surgeon Pioneer ——Barber–Surgeons

In the Middle Ages, cutting the human body was considered taboo. Therefore, surgical procedures such as tooth extractions, neoplasm excisions, and amputations were performed by barbers, who already possessed the necessary cutting tools.



(By Jacobus de Cessoli, 1408)

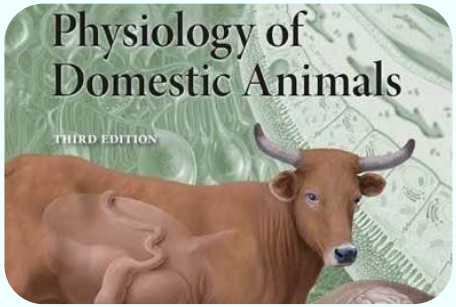
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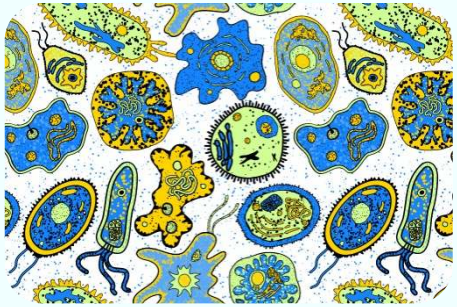
Anatomy



Pharmacology



Physiology



Microbiology



Presurgical considerations



Presurgical considerations

- **Evaluation of the patient**
- Surgical judgement
- Principle of asepsis and antisepsis
- Postoperative care

Evaluation of the Patient

Before a surgical procedure, a **physical examination** is generally indicated. This applies to both emergency and elective surgery. Exactly where to draw the line on laboratory tests is largely a matter of judgement on the part of the surgeon. However, additional tests will be necessary, such as:

Evaluation of the Patient

- Electrolyte measurement for right-sided abomasal diseases of the dairy cow
- Urinalysis in the dairy cow to evaluated the presence of ketosis
- Full chemistry panels when there are age or systemic considerations
- Measurement of blood urea nitrogen (BUN) and creatinine if urinary problems are suspected
- Analysis of peritoneal fluid prior to laparotomy for horses with colic
- Analysis of packed cell volume (PCV) and total protein as the foal younger than 4 years old

Tests for diagnosis of ketosis





Evaluation of the Patient

If any laboratory parameters are abnormal, the underlying causes should be investigated and efforts made to correct them. The owner should be made aware of any problems prior to subjecting the animal to surgery.

Medical records should be kept at all times. It should become and essential part of the procedure for horses and cattle (including herd records), even the procedure conduct on farm.

Presurgical considerations

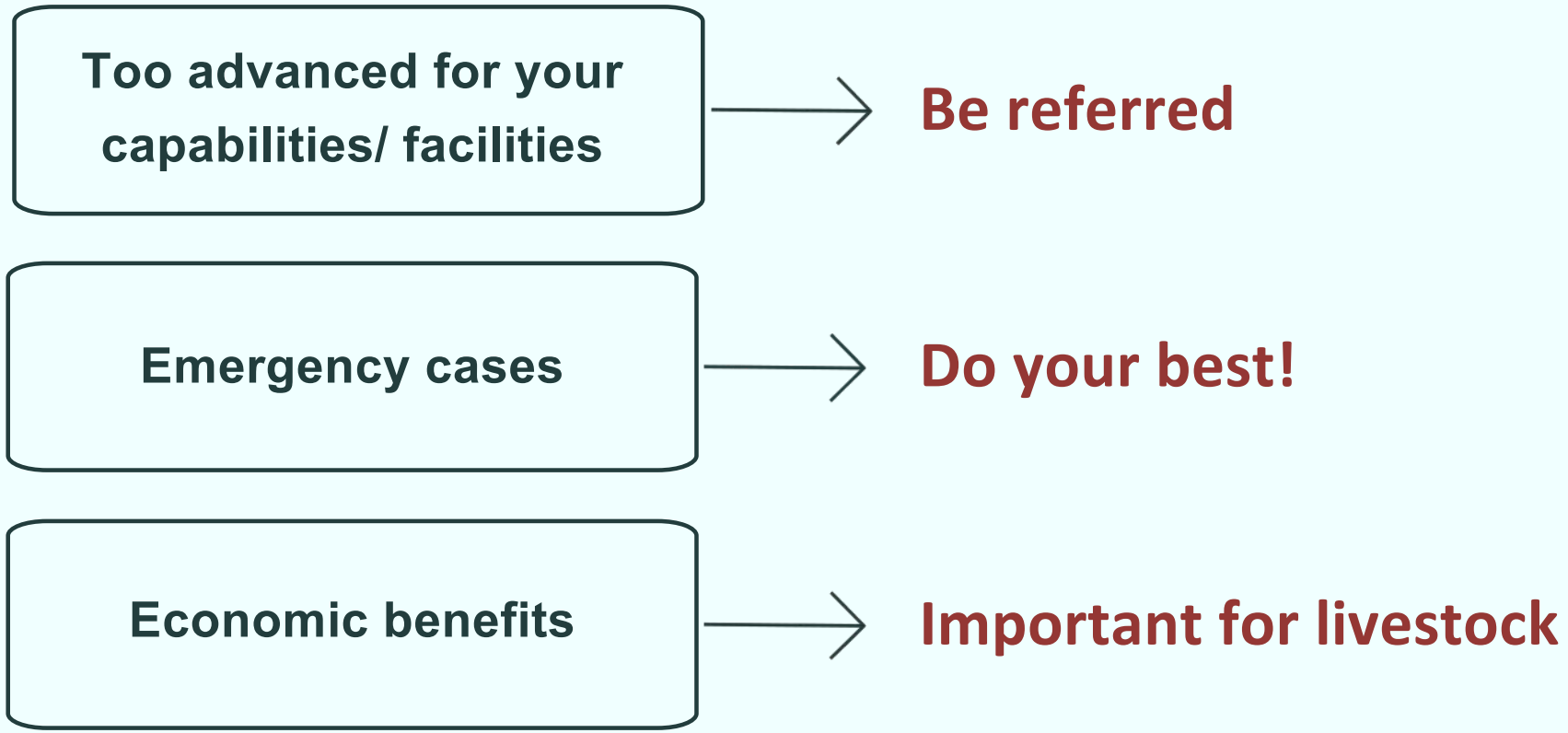
- Evaluation of the patient
- **Surgical judgement**
- Principle of asepsis and antisepsis
- Postoperative care

Surgical Judgement

As part of surgical judgement, the surgeon must ask the following questions:

- Is the surgery necessary?
- What would happen if the surgery were not performs?
- Is the procedure within the capabilities of the surgeon, the facilities, and the technical help?

Surgical Judgement



Surgical Judgement

Many of the of the surgery procedures for large animal (cattle, goat and sheep, pigs) can be done “on the farm.” However, some procedures, such as arthrotomy for removal of chip fractures of the carpal and sesamoid bones in horses, should be done in a dust-free operating theater.

Presurgical considerations

- Evaluation of the patient
- Surgical judgement
- **Principle of asepsis and antisepsis**
- Postoperative care

Principles of Asepsis and Antisepsis

There are 4 main determinants for a surgical site infection: host defense, physiologic derangement, bacterial contamination risk at surgery, and prolonged surgical time. Other factors that impact infection of deep structures and organs include hypoalbuminemia and a prior operation.

Principles of Asepsis and Antisepsis

Control methods include aseptic surgical practices as well as identification of the high-risk patient, correction of systemic imbalance prior to surgery, and the proper use of prophylactic antibiotics.

Principles of Asepsis and Antisepsis

Ideal standard may be unattainable “on the farm,” one should always strive for the highest possible standard; otherwise, the final standard of practice maybe so low that the well-being of the patient is at risk, not to mention the reputation of the veterinarian as a surgeon.

Surgical classifications

Classification	Description	Examples
Clean	GI, urinary, or respiratory tract is not entered	Arthrotomy
Clean-contaminated	GI, urinary, or respiratory tract is entered. There is no spillage of contaminated contents.	Abomasopexy for displacement abomasum
Contaminated-dirty	Gross spillage of contaminated body contents or acute inflammatory occurs.	Wounds, abscesses, devitalized bowl, rumenostomy, rumenotomy


Contaminated-dirty surgery
瘤胃切開術取出瘤胃內異物

Principles of Asepsis and Antisepsis

Once the surgeon has categorized the surgical procedure, appropriate precautions to avoid postoperative infection can be determined. In all cases, however, **the surgical site is prepared properly**, including clipping and aseptic scrubbing. Whatever category of surgery is performed, clean clothing should be worn.

Principles of Asepsis and Antisepsis

If the surgeon decides that antibiotics are indicated, special attention should be given to selecting the type od antimicrobial drug, the dosage, and the duration of use. For maximum benefit, antimicrobial should be administered prophylactically prior to surgery and, at the latest, during surgery. Antibiotics should never be used to cover flaws in surgical technique.

Presurgical considerations

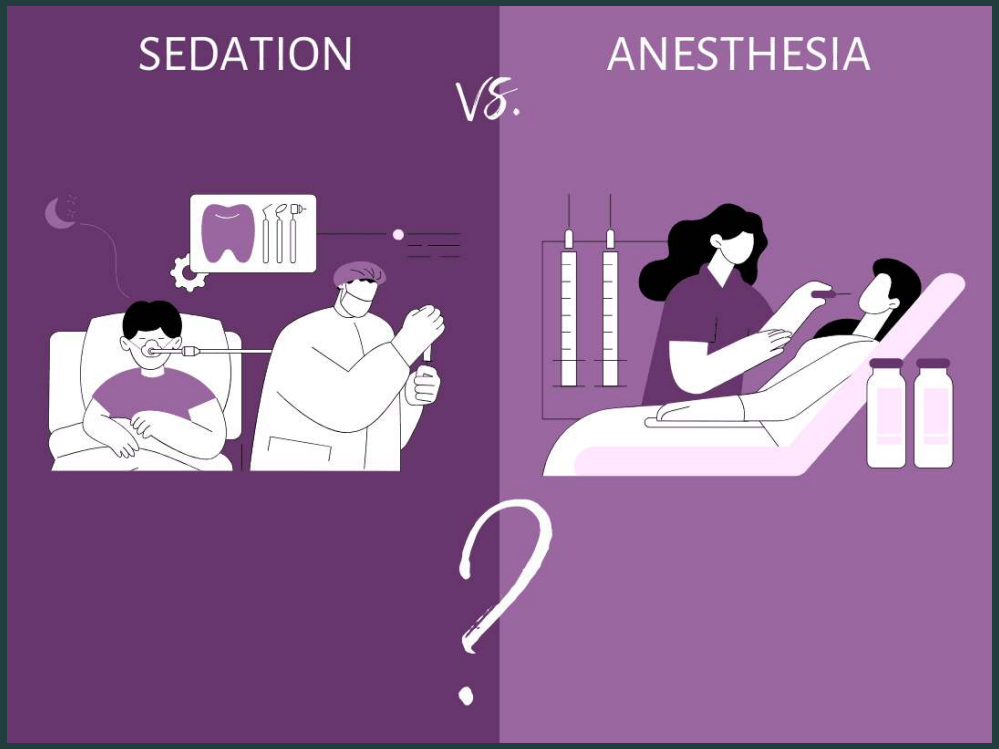
- Evaluation of the patient
- Surgical judgement
- Principle of asepsis and antisepsis
- **Postoperative care**

Postoperative care

- Postoperative medication
- Postoperative feeding and management
- Postoperative follow-up



Anesthesia and Sedation

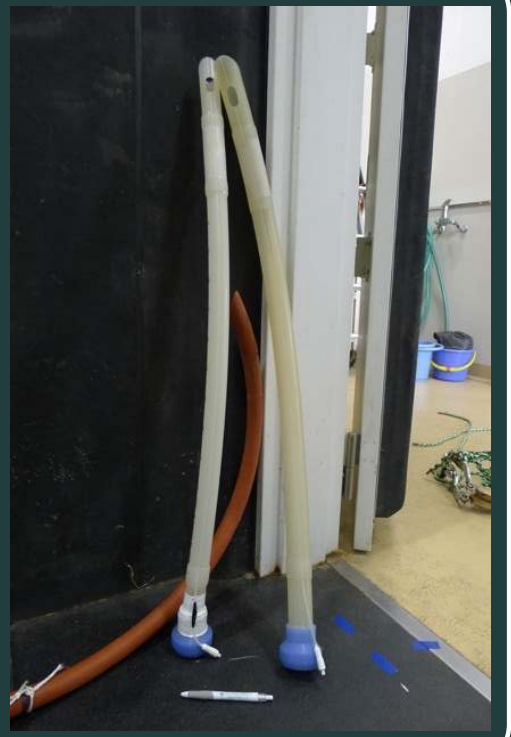


Anesthesia and Sedation

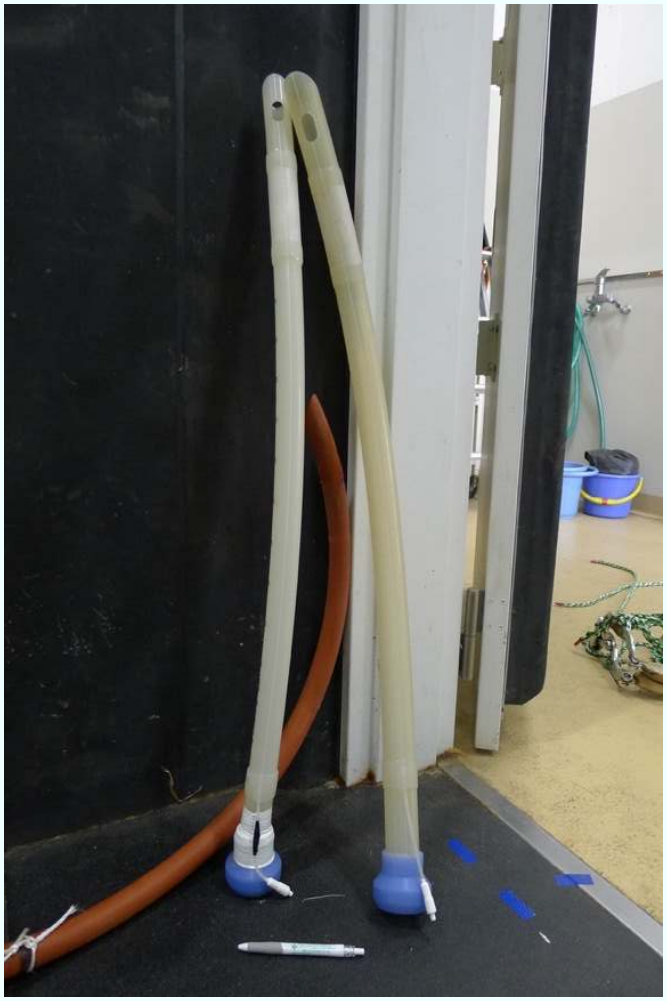
Cattle are ruminants with a very large rumen in their abdominal cavity. Therefore, prolonged recumbency under general anesthesia can easily cause bloating (also happened postoperatively), regurgitaion, and aspiration pneumonia. It can also interfere with diaphragmatic movement, leading to hypoventilation, hypoxia, hypercarbia, and respiratory acidosis.

Anesthesia and Sedation

Fasting ruminal patients prior to anesthesia and placement of a cuffed endotracheal tube will minimized the risk of regurgitation and subsequent aspiration during recumbency. Oro-ruminal tube may be placed to facilitate displacement of gas and liquid contents to reduce bloat.







Anesthesia and Sedation

Feed should need be withheld for 12–24 hours and water withheld overnight prior to surgery. Feed and water should be withheld from sheep and goats 12–24 and 6–12 hours, respectively, before surgery.

In calves, deep sedation can be used for chemical restraint, allowing veterinarians to perform surgery.

Sedation

Use of α-2 adrenoceptor agonist, such as xylazine hydrochloride, detomidine, medetomidine, dexmedetomidine, and romifidine is common as these drugs provide both sedation and analgesia. They have side effects including a drug-, dose-, and route-dependent period of hypertension (followed by hypotension), heart rate ↓, cardiac output ↓, ruminal atony and ileus.

Sedation

Reports indicate that xylazine has oxytocin-like effects in early-pregnancy cows, causing **uterine contractions**; therefore, pregnant cows should use this drug with special caution.

α-2 agonist drugs can be used in combination with opioids (ex. butorphanol) to reduce sudden responsiveness in some animals.

Sedation

Use of α-2 adrenoceptor agonist, such as xylazine. The dosage for casting agent or standing sedation in cattle:

Cattle: 0.02–0.1 mg/kg (IV)
0.11–0.22 mg/kg (IM)
0.015–0.025 mg/kg (IM, cows without recumbency)



Sedation

The effects of the sedative xylazine can be suppressed by antagonists such as yohimbine (0.12 mg/kg IV) or tolazoline (2–4 mg/kg IV). Side effects of these drugs include hypertension and arrhythmia. Clinically, tolazoline usually takes effect faster than yohimbine.

Infiltration / Regional analgesia

Regional analgesia results form desensitization of sensory nerve to a given area. This may be performed by infiltration into the desired location of by blocking sensory nerve innervating a region. Depending on the required duration of anesthesia, local anesthetic agents including lidocaine (short onset and duration), mepivacaine (rapid onset with intermediate duration, bupivacaine (long onset and duration) may be used.

Infiltration / Regional analgesia

Many active ingredients (core structures) of anesthetics are organic weak bases, which are poorly water-soluble and unstable. Converting them into hydrochloride salts (HCl) improves water solubility, enhances absorption and bioavailability, and stabilizes the chemical structure (extending shelf life). However, because these aqueous solutions are slightly acidic, they can cause temporary pain and a burning sensation at the injection site.

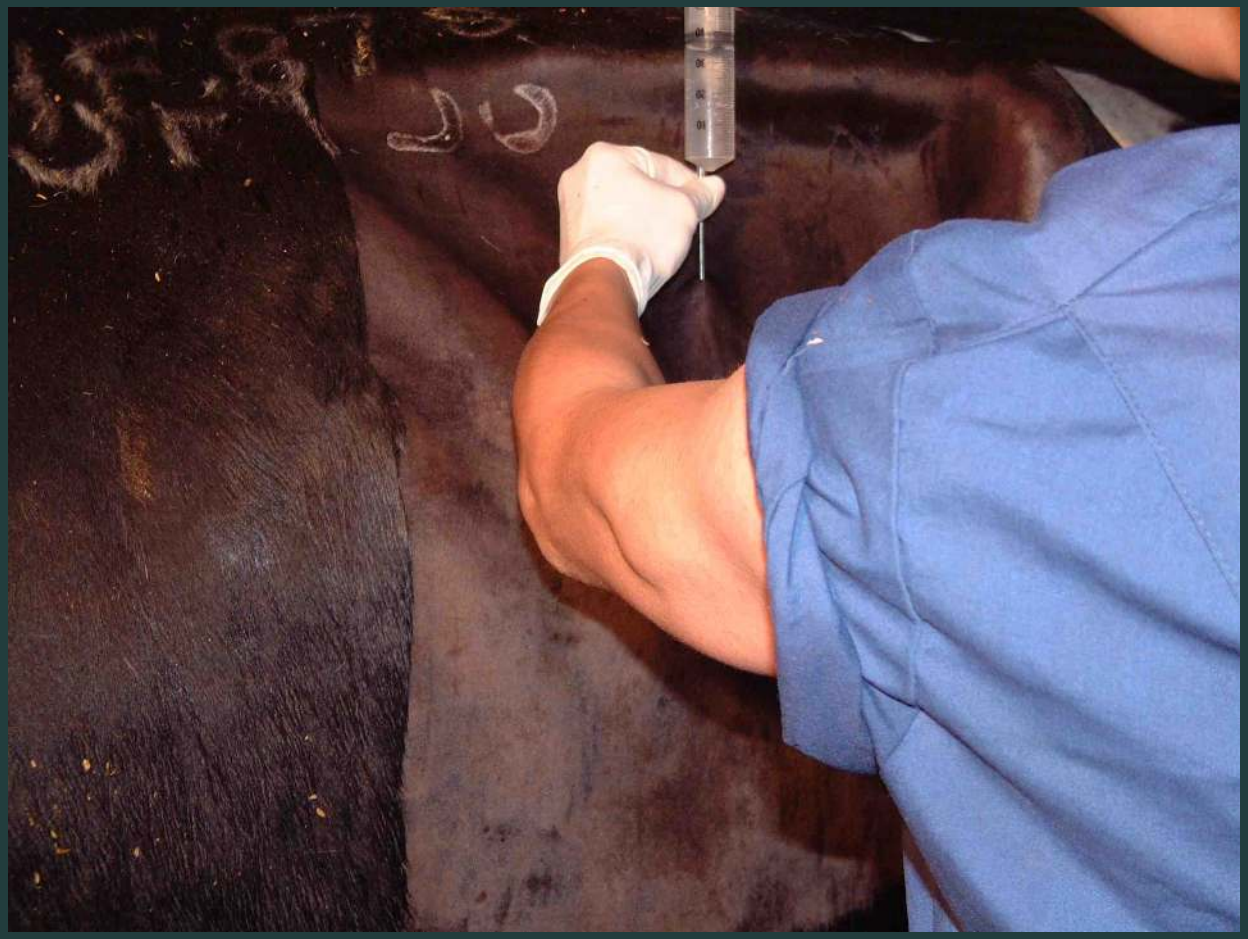
Infiltration / Regional analgesia

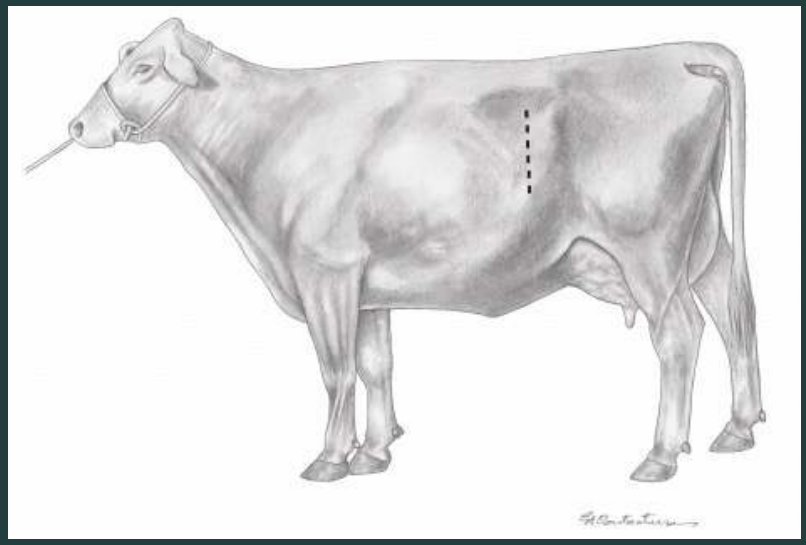
Due to cardiovascular toxicity with vascular absorption, bupivacaine use is usually limited to epidural and perineural administration. Lidocaine and mepivacaine may be used by any route.

Use of Lidocaine

The safe total dose of lidocaine is usually 3–5 mg/kg (90–150 mL of 2% lidocaine/600 kg cattle). Do not exceed the dose at one time. Lidocaine takes effect about 5–10 minutes after injection, and the analgesic effect lasts for about 90–180 minutes. Clinically, about 100 mL of 2% lidocaine is sufficient for general surgery on an adult cow.

Linear Infiltration Anesthesia







Infiltration needle
Dr. Wille's model, with side openings and small outlet,
140 mm long, 1,8 mm Ø,stainless

Infiltration Anesthesia

The skin and subcutis are infiltrating first and then the deeper layers of muscles and peritoneum. Significant amounts of local anesthetic inject into the peritoneal cavity should be avoided as rapid vascular absorption can result in toxicity. Injection should be made in straight lines. Although convenient, the infiltration of the analgesia agent into the incision line causes edema in the tissues and may affect wound healing.



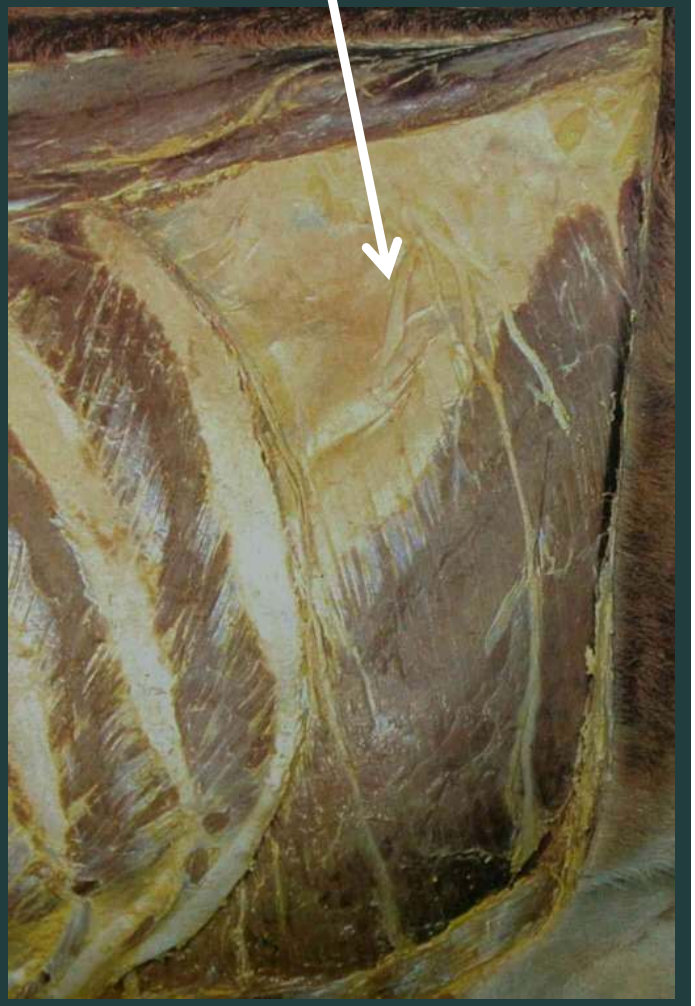
external oblique



internal oblique



Transversus abdominis





Infiltration Anesthesia

Inject approximately 3–5 mL of lidocaine at the uppermost point of the incision using an 18 G needle. Switch to a 15 G, 14-cm long needle. Slightly elevate the skin and advance the long needle vertically downward into the subcutis until the entire needle is submerged. Slowly inject the drug while concurrently retracting the needle to ensure a linear distribution along the incision site. Total volume of anesthetic used per layer should be about 25–30 mL.

Infiltration Anesthesia

Upon completion of subcutaneous anesthesia, hold the long needle horizontally and insert it approximately 1–1.5 cm into the muscular layer. Then, advance the needle vertically downward through the muscle layer until the entire needle is submerged. Inject the anesthetic while concurrently withdrawing the needle, using the same technique described above.

Infiltration Anesthesia

Two muscular layers must be anesthetized: the first layer can usually be confirmed as the external abdominal oblique muscle, while the second layer is less distinguishable and may include the internal abdominal oblique, transversus abdominis, and peritoneum.



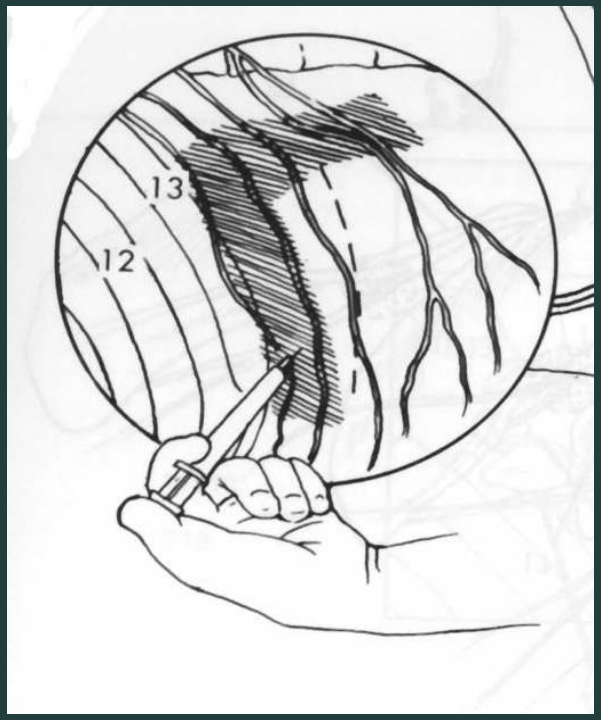
Inverted L Block



https://veteriankey.com/local-and-regional-anesthesia-for-urogenital-surgery/

Administer the drug along a horizontal line approximately 10 cm ventral to the lumbar transverse processes, and a vertical line 5–10 cm posterior to the last rib. Along both the horizontal and vertical paths, perform anesthetic blockade across one subcutaneous layer and two muscular layers.

Inverted L Block



This method primarily desensitizes the areas innervated by the 13th thoracic nerve (T13) and the 1st and 2nd lumbar nerves (L1 & L2). It is applicable for performing either a Caesarean section or a rumen fistulation.